Answer Sketch for 431 Quiz 2

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```
library(knitr)
library(janitor)
library(mosaic)
library(naniar)
library(ggrepel)
library(car)
library(magrittr)
library(patchwork)
library(vcd)
library(Epi)
library(Hmisc)
library(broom)
library(tidyverse)

opts_chunk$set(comment = NA)
theme_set(theme_bw())

source("data/Love-boost.R")

survA <- read_csv("data/survA.csv")
```

Setup for Questions 1-15

Questions 1-15 all refer to the same data set (the `survA.csv` data) which I have provided to you in our Shared Google Drive in the data subfolder within the Quiz 2 folder.

Consider the `survA.csv` file Dr. Love has provided. The file contains (lightly edited) responses to nine items that I have asked of students during the first week of 431, in each of the past seven years. The `survA.csv` file also includes a subject identification code called `student`, as well as the `year` (which ranges from 2014 through 2020) in which the survey was given. Ingest the data into R and complete analyses to address Questions 1-15.

- Note that each of the first 15 questions contains two parts.

1 Question 1

The `height_in` variable stores responses to the item "what is your height?" that have been converted to measurements in inches.

a. How many of the students who responded to the `height_in` item gave responses that fell within two standard deviations of the sample mean?

b. What percentage of students with a response to `height_in` would we expect to see in your count in Question 1a if the Normal distribution was a perfect model for the distribution? Round your answer to zero decimal places.

1.1 Answer 1a is 367. Answer 1b is 95.

```
favstats(~ height_in, data = survA)
```

```
 min Q1 median Q3 max    mean       sd   n missing
  57 64     67 70  78 67.1164 3.734873 378       4
```

We have 378 responses to the `height_in` item, with sample mean 67.12 and sample standard deviation 3.73, so the values that make up the range of interest are (59.66, 74.58) which means all values between 60 and 74, inclusive, since all of the responses are integer values. This turns out to be 367 of the 378 responses, or 97.1%.

```
survA %>% count(height_in > 59.66 & height_in < 74.58)
```

```
# A tibble: 3 x 2
  `height_in > 59.66 & height_in < 74.58`     n
  <lgl>                                   <int>
1 FALSE                                      11
2 TRUE                                      367
3 NA                                          4
```

If the Normal distribution was a perfect fit, based on the Empirical Rules, we'd expect approximately 95% (technically, 95.45%) of `height_in` values to be in this range, which we can verify with the following code.

```
pnorm(2, mean = 0, sd = 1) - pnorm(-2, mean = 0, sd = 1)
```

```
[1] 0.9544997
```

2 Question 2

The `lastsleep` variable contains responses to the item "How many hours did you sleep last night?"

a. In which `year` did the sample of `lastsleep` values have the smallest mean number of hours?
b. After rounding to one decimal place, what was the sample mean of `lastsleep` in the `year` you identified in Question 2a?

2.1 Answer 2a is 2020. Answer 2b is 6.6 hours.

```
favstats(lastsleep ~ year, data = survA) %>%
  select(year, mean) %>%
  arrange(mean)
```

```
  year     mean
1 2020 6.571642
2 2017 6.708333
3 2018 6.784314
4 2014 6.902439
5 2016 7.002698
6 2019 7.113333
7 2015 7.316327
```

3 Question 3

The most common response to the item "What is your favorite color?", which was gathered in the `favcolor` variable was blue, mentioned by nearly half of the respondents. The second most common response was green.

a. What color was the third most common choice, and how many subjects gave that response?
b. What color was the fourth most common choice, and how many subjects gave that response?

3.1 Answer 3a and 3b are purple and red which were selected by 40 subjects each.

```
survA %>% tabyl(favcolor) %>% arrange(desc(n))
```

```
 favcolor   n     percent valid_percent
     blue 167 0.437172775   0.446524064
    green  58 0.151832461   0.155080214
   purple  40 0.104712042   0.106951872
      red  40 0.104712042   0.106951872
     pink  13 0.034031414   0.034759358
   orange  12 0.031413613   0.032085561
   yellow  11 0.028795812   0.029411765
     gray  10 0.026178010   0.026737968
    black   8 0.020942408   0.021390374
    white   8 0.020942408   0.021390374
     <NA>   8 0.020942408            NA
    brown   5 0.013089005   0.013368984
   silver   2 0.005235602   0.005347594
```

4 Question 4

All subjects in the `survA` file were assigned a `student` code and `year` value by Dr. Love. The other nine variables were collected from their responses to the items on the survey.

a. Which of the nine variables (excluding `student` and `year`) provided has the largest number of missing values?
b. How many values are missing in the variable you identified in Question 4a?

4.1 Answer 4a is `favcolor`. Answer 4b is 8.

```
miss_var_summary(survA)
```

```
# A tibble: 11 x 3
   variable   n_miss pct_miss
   <chr>       <int>    <dbl>
 1 favcolor        8    2.09
 2 haircut         6    1.57
 3 height_in       4    1.05
 4 english         3    0.785
 5 alone           3    0.785
 6 lastsleep       3    0.785
 7 statfuture      2    0.524
```

```
 8 lecture         2    0.524
 9 smoker          1    0.262
10 student         0    0
11 year            0    0
```

5 Question 5

a. Identify the `student` code for the subject with the largest number of missing values.
b. For which variables (excluding `student` and `year`) has the subject you identified in Question 5a provided data?

5.1 Answer 5a is 201432. Answer 5b is `english` and `smoker`.

```r
miss_case_summary(survA) %>% head(2)
```

```
# A tibble: 2 x 3
   case n_miss pct_miss
  <int>  <int>    <dbl>
1    32      7     63.6
2   260      6     54.5
```

```r
slice(survA, 32)
```

```
# A tibble: 1 x 11
  student english smoker statfuture haircut lecture alone height_in favcolor
    <dbl> <chr>    <dbl>      <dbl>   <dbl>   <dbl> <dbl>     <dbl> <chr>
1  201432 English      1         NA      NA      NA    NA        NA <NA>
# ... with 2 more variables: lastsleep <dbl>, year <dbl>
```

6 Question 6

Ava and Ivy are two analysts working separately, who wish to compare the `haircut` prices across the survey years using an appropriate overall test of whether the average (mean or pseudo-median) `haircut` values are detectably different by year, using a 90% confidence level. Each analyst assumes all missing `haircut` prices are missing completely at random. Ava also assumes that each year's `haircut` prices follow a Normal distribution with similar variance across years. Ivy, on the other hand, assumes that the `haircut` prices are symmetric, but that they do not follow a Normal distribution due to problems in the tails of the distributions.

a. State the p values (rounded to three decimal places) for each of the two analysts (Ava and Ivy) under the assumptions stated above.
b. Which of the analysts will declare a statistically detectable difference between the average `haircut` prices across years?
    a. Ava alone
    b. Ivy alone
    c. Both Ava and Ivy
    d. Neither Ava nor Ivy

6.1 Answer 6a is Ava (.190) and Ivy (.096). Answer 6b is choice b (Ivy only.)

It seems as though Ava will do an ANOVA, while Ivy will do a Kruskal-Wallis test.

```r
ava <- anova(lm(haircut ~ factor(year), data = survA))

ava
```

```
Analysis of Variance Table

Response: haircut
              Df Sum Sq Mean Sq F value Pr(>F)
factor(year)   6   6122 1020.40  1.4614 0.1904
Residuals    369 257652  698.24
```

- Note that identifying `year` as a factor was critical. Failing to do so would force the years to show a linear effect on haircut, and would lead to a very different p value, which was a common choice in the responses given by students this year.

```r
ava_incorrect <- anova(lm(haircut ~ year, data = survA))
ava_incorrect
```

```
Analysis of Variance Table

Response: haircut
           Df Sum Sq Mean Sq F value  Pr(>F)
year        1   2552 2552.46  3.6544 0.05668 .
Residuals 374 261222  698.45
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
ivy <- kruskal.test(haircut ~ factor(year), data = survA)

ivy
```

```
    Kruskal-Wallis rank sum test

data:  haircut by factor(year)
Kruskal-Wallis chi-squared = 10.755, df = 6, p-value = 0.09625
```

- Actually, R assumes when running a Kruskal-Wallis test that the `year` is a factor regardless of whether you specify this, so many people had problems with Ava's analysis, but not Ivy's, and thus received partial credit.
- In grading 6b, I decided to award credit based on the answer you provided in 6a.

7 Question 7

The `lecture` variable describes agreement (1 = Strongly Disagree to 5 = Strongly Agree) with the statement "I prefer to learn from lectures over learning from activities." The `alone` variable described agreement (on the same 1-5 scale) with the statement "I prefer to work on projects alone over working on a team.") Create a 5x5 table comparing lecture to alone (with `lecture` in the rows and `alone` in the columns) for the subjects in the data who have complete information on the `lecture` and `alone` variables, and also have a value in the `height_in` variable.

a. Following the Question 7 instructions to build a table, you should find three cells which contain just one subject. Identify the `student` code for the tallest of those three subjects.
b. How many of the three subjects who are the only members of their cell in the table from Question 7a agreed more with the `lecture` statement than with the `alone` statement?

7.1 Answer 7a is 201934. Answer 7b is 2.

```
q7 <- survA %>% select(student, lecture, alone, height_in) %>%
  filter(complete.cases(.))

q7 %>% tabyl(lecture, alone)
```

```
 lecture  1  2  3  4 5
       1 10  7  6  2 1
       2  9 28 43 23 3
       3 13 40 47 39 9
       4  1 13 26 30 6
       5  2  2  8  1 8
```

The three subjects are identified below. The one with the largest `height_in` is student 201934. 2 of the 3 subjects have a higher `lecture` response than their `alone` response.

```
q7 %>% filter(lecture == 1 & alone == 5)
```

```
# A tibble: 1 x 4
  student lecture alone height_in
    <dbl>   <dbl> <dbl>     <dbl>
1  201934       1     5        70
```

```
q7 %>% filter(lecture == 4 & alone == 1)
```

```
# A tibble: 1 x 4
  student lecture alone height_in
    <dbl>   <dbl> <dbl>     <dbl>
1  201633       4     1        68
```

```
q7 %>% filter(lecture == 5 & alone == 4)
```

```
# A tibble: 1 x 4
  student lecture alone height_in
    <dbl>   <dbl> <dbl>     <dbl>
1  202049       5     4        61
```

8 Question 8

Now use R to combine the responses 1 and 2 and 4 and 5 for the `lecture` and `alone` variables discussed in Question 7, and then create a new contingency table with `lecture` in the rows and `alone` in the columns. Your new table should have 3 rows and 3 columns. Now, use an appropriate hypothesis test to assess, with 90% confidence, whether the rows in your new table are independent of the columns.

a. What type of hypothesis test did you use to address this question?
    a. A test based on the t distribution
    b. A test based on the chi-square distribution
    c. A test based on a bootstrap
    d. A test based on the F distribution
    e. None of these
b. After rounding to five decimal places, what p value do you obtain from your test in Question 8a?

8.1 Answer 8a is choice b (a chi-square test). Answer 8b is 0.00054

The options for question 8a are:

a. A test based on the t distribution
b. A test based on the chi-square distribution
c. A test based on a bootstrap
d. A test based on the F distribution
e. None of these

and we need a Pearson chi-square test, so that's choice b.

```
q8 <- q7 %>%
  mutate(lec_new = fct_recode(factor(lecture),
                              "Low" = "1",
                              "Low" = "2",
                              "Mid" = "3",
                              "High" = "4",
                              "High" = "5"),
         alo_new = fct_recode(factor(alone),
                              "Low" = "1",
                              "Low" = "2",
                              "Mid" = "3",
                              "High" = "4",
                              "High" = "5"))

q8 %$% table(lec_new, alo_new)
```

```
       alo_new
lec_new Low Mid High
   Low   54  49   29
   Mid   53  47   48
   High  18  34   45
```

```
chisq.test(q8 %$% table(lec_new, alo_new))
```

```
    Pearson's Chi-squared test

data:  q8 %$% table(lec_new, alo_new)
X-squared = 19.843, df = 4, p-value = 0.0005363
```

9 Question 9

Now consider the `english` variable, which specifies for each subject whether their most comfortable language is English, or some other language, and the `smoker` variable, which divides subjects into group 1 (non-smokers), or group 2 (former smoker), or group 3 (current smoker). For Question 9, including only the subjects with complete data on these two variables. Collapse the `smoker` responses of 2 and 3 into a single group, which we'll call the smokers for this question. Now, estimate the odds ratio for being a non-smoker for students who are most comfortable speaking in a language other than English, as compared to students who are most comfortable speaking in English.

a. Provide a point estimate for the odds ratio identified in Question 9, rounded to two decimal places.
b. Again rounding to two decimal places, specify a 90% confidence interval for the estimate you provided in Question 9a.

9.1 Answer 9a is 0.80, or 0.65 if you use a Bayesian adjustment. Answer 9b is (0.34, 1.90), or (0.31, 1.40) if you use a Bayesian adjustment.

First, let's create the table in the needed format.

```
q09 <- survA %>%
  select(student, english, smoker) %>%
  filter(complete.cases(.)) %>%
  mutate(smoke = fct_lump_n(factor(smoker), 1),
         smoke = fct_recode(smoke,
                            "Non-Smoker" = "1"),
         eng = fct_relevel(factor(english), "Not_English"))

q09 %>%
  tabyl(eng, smoke)
```

```
         eng Non-Smoker Other
 Not_English         68     5
     English        289    17
```

Now, we can use `twobytwo` from `Love-boost` or we can use `twoby2` from `Epi` to get the odds ratio estimates we need from the `Sample Odds Ratio` row in the table below. We also could have done this with or without a Bayesian adjustment.

9.1.1 `twobytwo` and no Bayesian adjustment

```
twobytwo(68, 5, 289, 17,
         "Not English", "English", "Non-Smoker", "Smoker",
         conf.level = 0.9)
```

```
2 by 2 table analysis:
-------------------------------------------------------
Outcome   : Non-Smoker
Comparing : Not English vs. English

            Non-Smoker Smoker    P(Non-Smoker) 90% conf. interval
Not English         68      5           0.9315    0.8639   0.9668
English            289     17           0.9444    0.9185   0.9624

                                     90% conf. interval
             Relative Risk:  0.9863    0.9317   1.0441
         Sample Odds Ratio:  0.8000    0.3366   1.9013
Conditional MLE Odds Ratio:  0.8005    0.3131   2.3396
    Probability difference: -0.0129   -0.0809   0.0303

             Exact P-value: 0.5893
        Asymptotic P-value: 0.6716
-------------------------------------------------------
```

9.1.2 `twoby2` including Bayesian adjustment

```
twoby2(q09 %$% table(eng, smoke) + 2, conf.level = 0.90)
```

```
2 by 2 table analysis:
-----------------------------------------------------
Outcome   : Non-Smoker
Comparing : Not_English vs. English

            Non-Smoker Other    P(Non-Smoker) 90% conf. interval
Not_English         70     7           0.9091    0.8390   0.9505
English            291    19           0.9387    0.9121   0.9576

                                   90% conf. interval
             Relative Risk:  0.9684    0.9085   1.0324
         Sample Odds Ratio:  0.6529    0.3055   1.3954
Conditional MLE Odds Ratio:  0.6537    0.2862   1.6162
    Probability difference: -0.0296   -0.1008   0.0191

             Exact P-value: 0.3210
        Asymptotic P-value: 0.3559
-----------------------------------------------------
```

- Note: several people computed the 95% CI instead, which would be (0.29, 2.24) without a Bayesian augmentation.
- Note also that several people calculated the odds ratio associated with English vs. Non-English, instead of the reverse. That would flip the odds ratio estimate to 1.25, with 90% confidence interval (0.53, 2.97) assuming no Bayesian adjustment. If so, you received half-credit for Question 9a, but full credit on Question 9b.

10 Question 10

There are 3 subjects in the `survA` data who are missing exactly two values across the entire data set.

a. Of the 3 subjects missing exactly two values, how many are missing `favcolor`?
b. How many subjects in the entire `survA` data set have no missing values at all?

10.1 Answer 10a is 1. Answer 10b is 364.

```r
miss_case_summary(survA)
```

```
# A tibble: 382 x 3
    case n_miss pct_miss
   <int>  <int>    <dbl>
 1    32      7    63.6
 2   260      6    54.5
 3   365      2    18.2
 4   379      2    18.2
 5   381      2    18.2
 6     9      1     9.09
 7    26      1     9.09
 8    27      1     9.09
 9    30      1     9.09
10    58      1     9.09
# ... with 372 more rows
```

```r
survA %>% slice(c(365, 379, 381))
```

```
# A tibble: 3 x 11
  student english smoker statfuture haircut lecture alone height_in favcolor
    <dbl> <chr>    <dbl>      <dbl>   <dbl>   <dbl> <dbl>     <dbl> <chr>
1  202050 English      1          6      NA       1    NA        60 blue
2  202064 <NA>        NA          5       0       2     2        63 yellow
3  202066 Not_En~      1          5      25       4     5        NA <NA>
# ... with 2 more variables: lastsleep <dbl>, year <dbl>
```

Of these three, only subject 202066 is missing `favcolor`.

```r
miss_case_table(survA)
```

```
# A tibble: 5 x 3
  n_miss_in_case n_cases pct_cases
           <int>   <int>     <dbl>
1              0     364    95.3
2              1      13     3.40
3              2       3     0.785
4              6       1     0.262
5              7       1     0.262
```

11 Question 11

Consider the responses to the item "How important do you think statistics will be in your future career?" as gathered in the `statfuture` variable. There were four years prior to 2020 where the class had no missing responses on this item. Use the data from those four years to estimate the proportion of subjects who responded with the value 7 (meaning extremely important), using the approach due to Agresti and Coull.

a. Rounding to three decimal places, specify a 90% confidence interval for the true value of the proportion described in Question 11.
b. Is the sample proportion of "7" values from the 2020 survey inside the interval you reported in Question 11a?
    a. Yes, it is inside the interval
    b. No, the sample proportion is lower than all values in the interval
    c. No, the sample proportion is higher than all values in the interval

11.1 Answer 11a: 90% CI is (0.495, 0.607). Answer 11b is choice b (No and lower)

```r
survA %>% tabyl(statfuture, year)
```

```
 statfuture 2014 2015 2016 2017 2018 2019 2020
          3    0    1    0    0    0    0    0
          4    1    1    1    0    0    0    2
          5    8   10    7    8    6    4    9
          6   14   18   19   10   14   14   28
          7   18   19   37   30   31   42   28
         NA    1    0    0    0    0    1    0
```

2015-2018 are the surveys with no missing responses on this item before 2020.

```r
q11 <- survA %>%
  select(student, statfuture, year) %>%
  filter(year %in% c(2015:2018))

q11 %>% count(statfuture == 7)
```

```
# A tibble: 2 x 2
  `statfuture == 7`     n
  <lgl>             <int>
1 FALSE                95
2 TRUE                117
```

117 out of (117 + 95 = 212) of the responses in 2015-18 were 7.

```r
mosaic::binom.test(x = 117, n = 212, conf.level = 0.90,
                   ci.method = "Agresti-Coull")
```

```

    Exact binomial test (Agresti-Coull CI)

data:  117 out of 212
number of successes = 117, number of trials = 212, p-value = 0.149
alternative hypothesis: true probability of success is not equal to 0.5
90 percent confidence interval:
 0.4954009 0.6070650
sample estimates:
probability of success
             0.5518868
```

How about 2020?

```r
survA %>% filter(year == 2020) %>%
  count(statfuture == 7)
```

```
# A tibble: 2 x 2
  `statfuture == 7`     n
  <lgl>             <int>
1 FALSE                39
2 TRUE                 28
```

Note that 28 / (28+39) = 0.418, which is lower than all values in the interval.

- Note also that if you calculated an incorrect interval in 11a, in grading 11b, I gave you credit only if your response was consistent with a correct calculation of 0.418, and the incorrect interval you specified in 11a.

12 Question 12

Fit two linear models to predict `haircut` price on the basis of `height_in` and `english`, in each case using only those subjects with complete data on each of those three variables, as follows.

- In model **m12A**, use only `height_in` to predict `haircut`.
- In model **m12B**, use both `height_in` and `english` to predict `haircut`.

a. Specify the values of BIC for models m12A and m12B, rounded to zero decimal places.
b. According to the AIC and BIC, which of the two models (m12A or m12B) shows stronger performance?
    a. Model m12A shows stronger performance on both AIC and BIC

b. Model m12B shows stronger performance on both AIC and BIC
c. Model m12A is better on AIC, but m12B is better on BIC
d. Model m12A is better on BIC, but m12B is better on AIC

12.1 Answer 12a is m12A: BIC = 3484. m12B: BIC = 3483. Answer 12b is choice b (m12B shows stronger performance using both AIC and BIC.)

```
q12 <- survA %>%
  select(student, english, height_in, haircut) %>%
  filter(complete.cases(.))

m12A <- lm(haircut ~ height_in, data = q12)
m12B <- lm(haircut ~ height_in + english, data = q12)

res_12 <- bind_rows(glance(m12A), glance(m12B)) %>%
  mutate(modelname = c("m12A", "m12B")) %>%
  select(modelname, AIC, BIC)

res_12
```

```
# A tibble: 2 x 3
  modelname   AIC   BIC
  <chr>     <dbl> <dbl>
1 m12A      3472. 3484.
2 m12B      3468. 3483.
```

Model `m12B` has lower AIC and BIC, and thus (slightly) better performance.

- Several people got 3518 for `m12A`, and that's just failing to filter to complete cases on the three specified variables, but you would still get 3483 for BIC in `m12B` so you should have received partial credit.
- If you got Question 12a wrong, but specified a BIC for each model, I gave you half credit on Question 12b for a response that went with your 12a response, but you couldn't get full credit on 12b.

13 Question 13

Using model m12B as described in Question 12, identify the standardized residuals for each subject, and identify the largest (in absolute value) of these standardized residuals.

a. Specify the `student` code for the subject with the largest (in absolute value) standardized residual in model m12B.
b. Specify the fitted and observed values of `haircut` for the subject you identified in Question 13a, in each case rounding to the nearest integer.

13.1 Answer 13a is 201532. Answer 13b: fitted = 25, observed = 210

```
aug13 <- augment(m12B, data = q12)

aug13 %>% arrange(desc(.std.resid)) %>%
  select(student, .std.resid, .fitted, haircut) %>%
  head(2)
```

```
# A tibble: 2 x 4
  student .std.resid .fitted haircut
    <dbl>      <dbl>   <dbl>   <dbl>
1  201532       7.21    25.0     210
2  202016       4.09    25.0     130
```

- Note that if you got 25 and 210 didn't label which was which, you needed to put them in the order I asked for them (fitted then observed) to get full credit.

14 Question 14

Consider again the subject you identified by their `student` code in Question 13.

a. Rounded to three decimal places, what is the Cook's distance (using model m12B) for the subject identified in Question 13?
b. Does Cook's distance identify that subject as influential on model m12B?
    a. Yes
    b. No
    c. It is impossible to tell from the available information

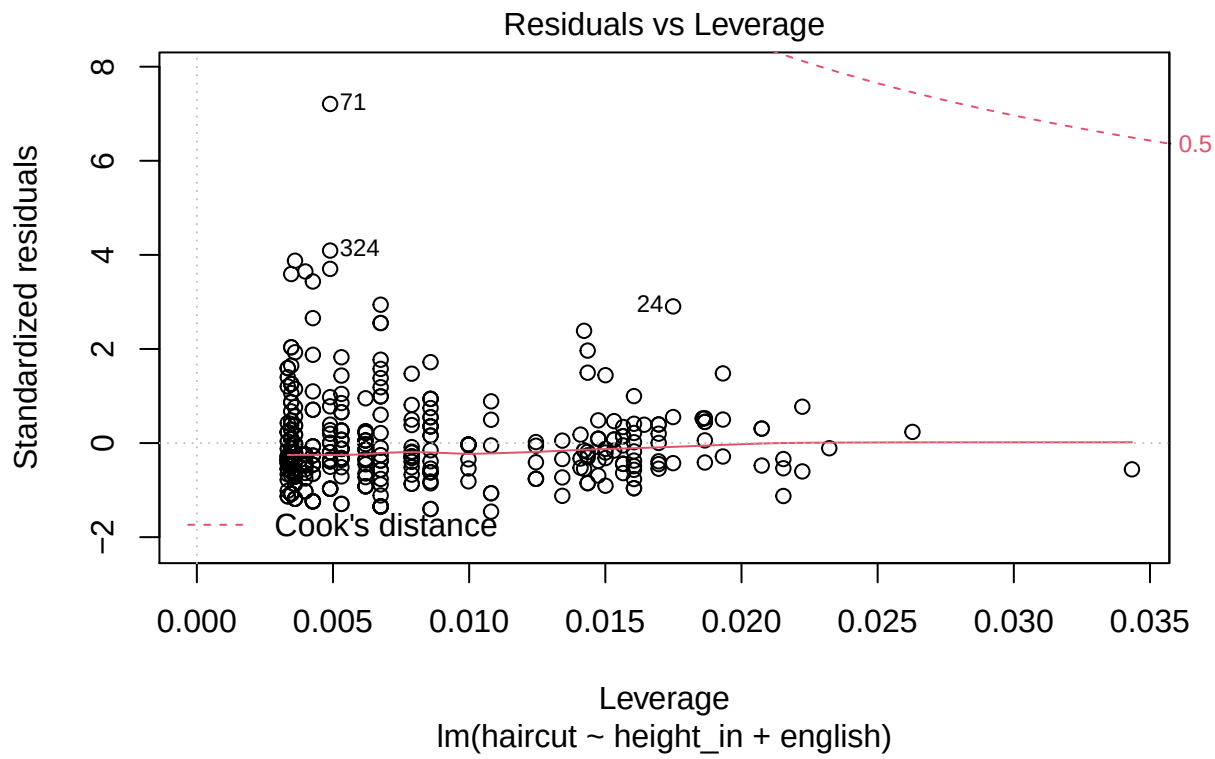
14.1 Answer 14a is 0.085. Answer 14b is choice b (No.)

```r
aug13 %>% arrange(desc(.std.resid)) %>%
  select(student, .std.resid, .cooksd) %>%
  head(2)
```

```
# A tibble: 2 x 3
  student .std.resid .cooksd
    <dbl>      <dbl>   <dbl>
1  201532       7.21  0.0852
2  202016       4.09  0.0275
```

The Cook's distance is less than 0.5 for this student, so no, it would not be identified as influential.

```r
plot(m12B, which = 5)
```



15 Question 15

Now, let's create a new model, called model **m_15**, where you rerun model m12B, but this time excluding the `student` you identified in Question 13.

a. Specify the residual standard error (rounded to 1 decimal place) and the R-squared value (expressed as a proportion, and rounded to 3 decimal places) for model m_15.
b. Consider an appropriate plot of the standardized regression residuals for model m15. Is there a problem with the assumption of Normality?
   a. Yes, because there is substantial skew in the distribution of the standardized residuals
   b. Yes, because the residuals are symmetric, but there are several problematic outliers
   c. Yes, because the residuals are symmetric, but their distribution is light-tailed compared to a Normal model
   d. No, because the residuals are well-approximated by a Normal distribution

15.1 Answer 15a: RSE (sigma) = 23.9, R^2 = 0.066. Answer 15b is choice a (Yes, there is substantial skew in the residual plot.)

```
q15 <- survA %>%
  select(student, english, height_in, haircut) %>%
  filter(complete.cases(.)) %>%
  filter(student != 201532)
```

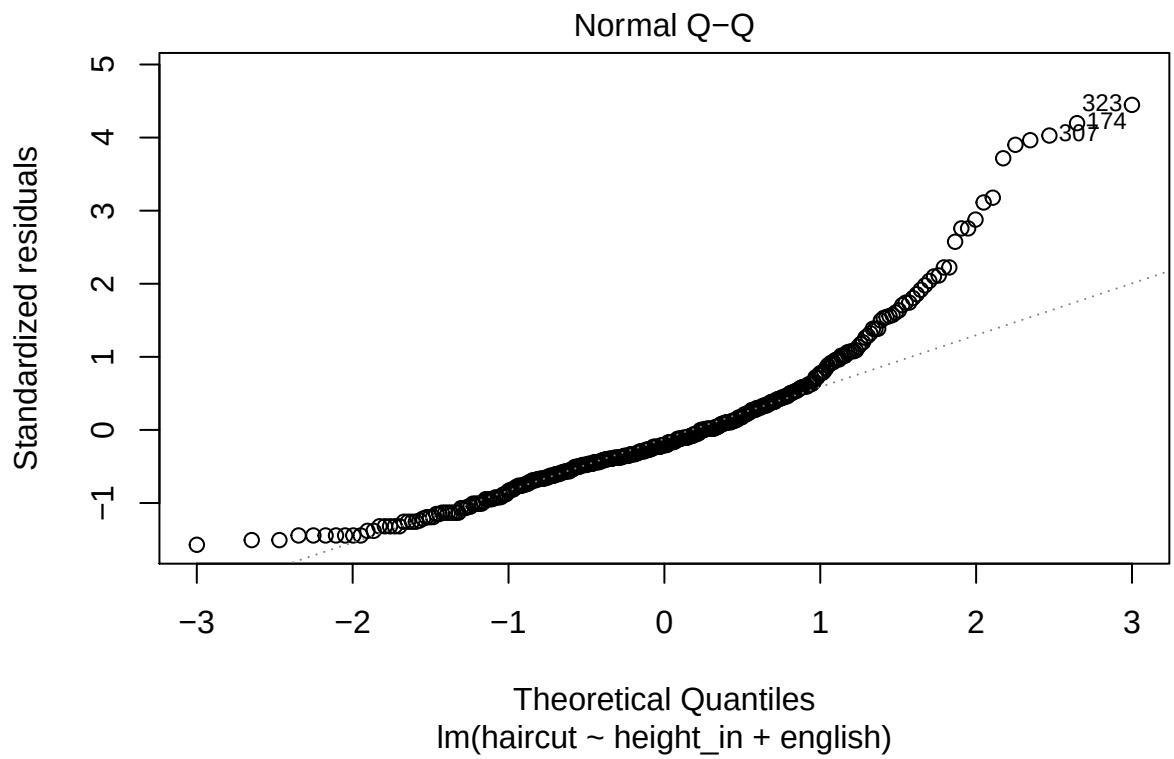
```r
m15 <- lm(haircut ~ height_in + english, data = q15)

glance(m15) %>% select(sigma, r.squared)
```

```
# A tibble: 1 x 2
  sigma r.squared
  <dbl>     <dbl>
1  23.9    0.0661
```

Here's the Q-Q plot of the regression's standardized residuals. There's a lot of skew, even after removing student 201532.

```r
plot(m15, which = 2)
```



16 Question 16 (4 points)

Data from a pilot study describe the pre-operative and post-operative creatinine clearance (in ml/minute) of six patients anesthetized with a new drug combination we want to test.

Subject	Pre-operative	Post-operative
1	110	137
2	101	93
3	71	99
4	73	91
5	133	112

Subject	Pre-operative	Post-operative
6	118	134

Suppose we are willing to assume that the results of this study are an appropriate pilot for a larger study we are designing that will also compare pre-operative and post-operative creatinine clearance rates. Suppose we want to maintain a two-sided 5% significance level. Suppose that an effect that is at least half the effect size seen in the pilot study will be our minimum standard for clinical importance. We also assume that the pilot's standard deviation estimates are 25% too low, so we'll multiply them by 1.25 in developing our design.

As a hint, consider the following output:

```
subj <- 1:6
pre <- c(110, 101, 71, 73, 133, 118)
post <- c(137, 93, 99, 91, 112, 134)

pilot <- tibble(subj, pre, post)

favstats(~ pre, data = pilot)
```

```
 min Q1 median  Q3 max mean       sd n missing
  71 80  105.5 116 133  101 24.81129 6       0
```

```
favstats(~ post, data = pilot)
```

```
 min   Q1 median    Q3 max mean       sd n missing
  91 94.5  105.5 128.5 137  111 20.36664 6       0
```

```
favstats(~ pre-post, data = pilot)
```

```
 min     Q1 median Q3 max mean    sd n missing
 -28 -24.75    -17  2  21  -10 19.99 6       0
```

Under the conditions specified above, you have been asked to determine the smallest number of subjects we must include in our new study to obtain 90% power to detect the minimum clinically important effect.

a. What is the smallest number of subjects the new study will have to include?
b. Provide your R code (a single line will suffice) to justify your response to Question 16a.

16.1 Answer 16a is 265 subjects. Answer 16b is the following R code.

```
power.t.test(delta = -5, sd = 19.99*1.25,
             sig.level = 0.05, power = 0.90, type = "paired")
```

```
     Paired t test power calculation

              n = 264.3512
          delta = 5
             sd = 24.9875
      sig.level = 0.05
          power = 0.9
    alternative = two.sided

NOTE: n is number of *pairs*, sd is std.dev. of *differences* within pairs
```

- This is a paired samples study.

- We're comparing means, so we need `power.t.test`.
- The observed effect size was 10, so we need at least half that for our `delta`. (Note that whether you use -5 or +5 for delta has no effect.)
- The standard deviation in the sample set of differences was 20 (or technically 19.99). Increasing that by 20% means 19.99(1.25) = 24.9875. (If you use 25 instead you get the same conclusion.)
- We want a 95% confidence level, so we need sig.level = 0.05
- We want 90% power, so power = 0.90.

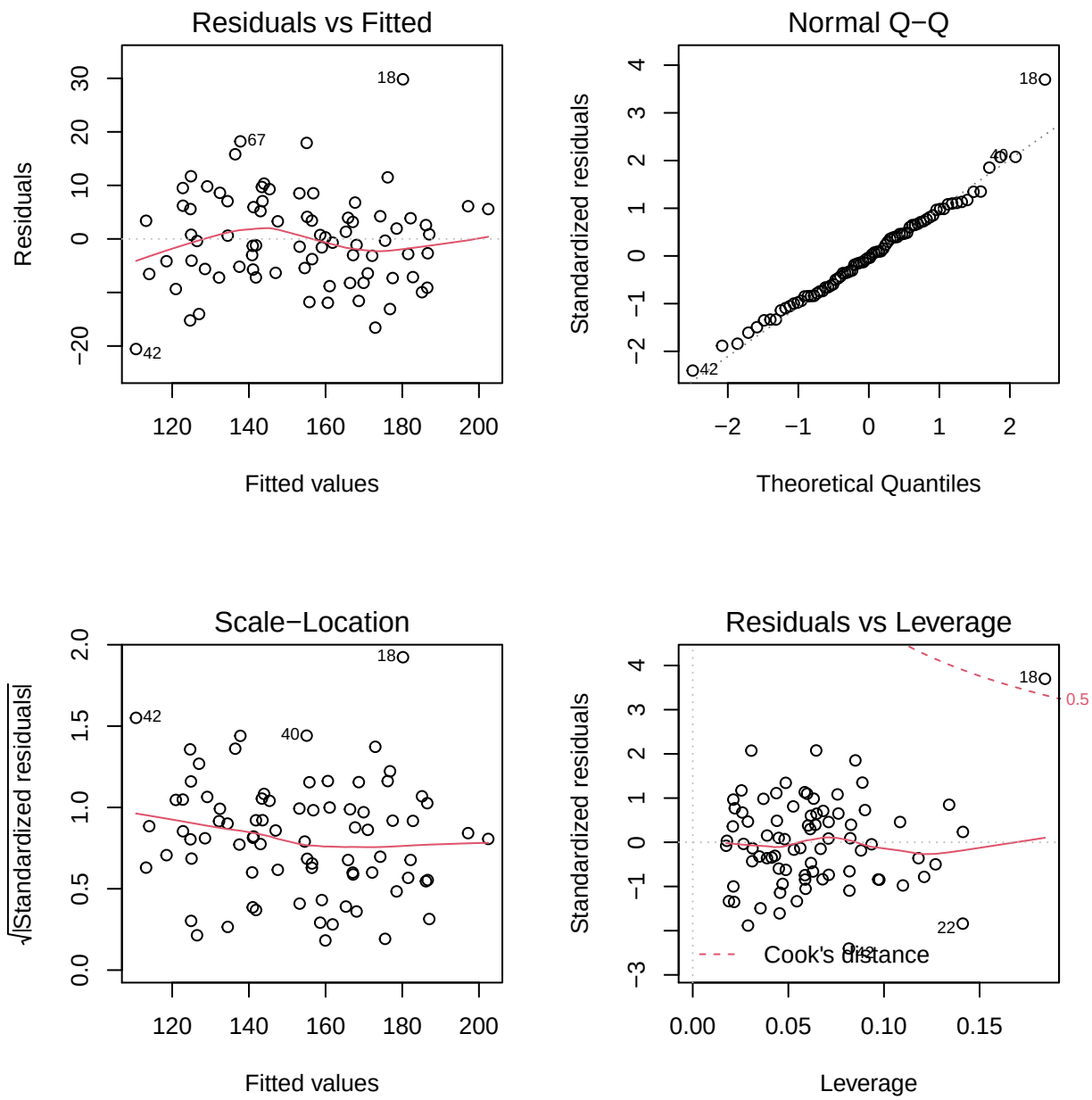
We obtain a paired t test with 265 observations (always round up to ensure we have the sample size necessary) and we realize that while each of those 265 subjects will be measured twice, that doesn't make them new subjects, so 265 is the minimum number of subjects.

Setup for Question 17

In this question, we're going to look at three sets of regression residual plots, for three separate models, called Model 1, Model 2, and Model 3. The three models estimate different outcomes.

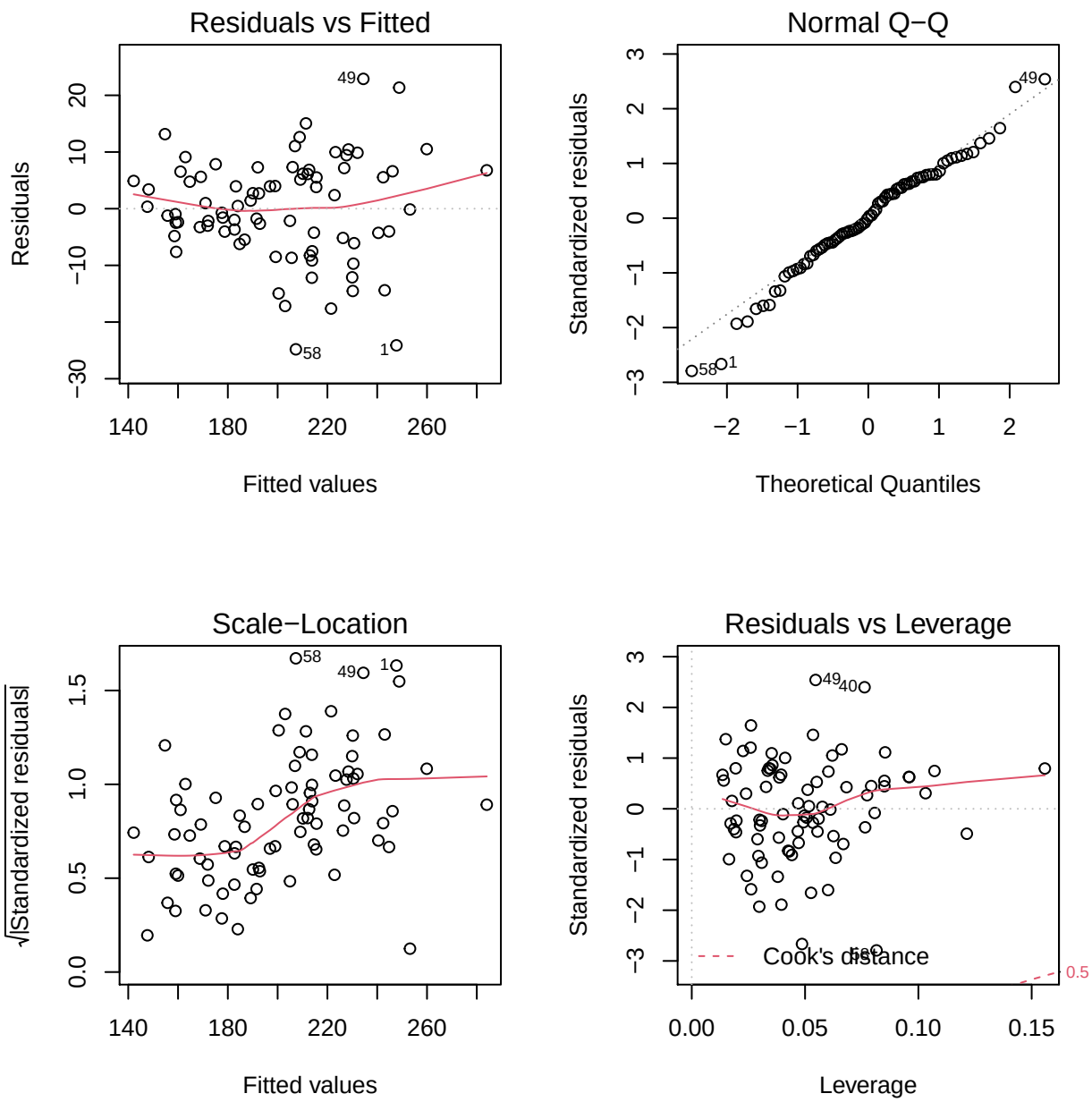
Question 17: Residual Plots - Model 1

Here are the plots for model 1, which includes 80 subjects.


Residuals vs Fitted
Residuals
Fitted values
18
67
42
Normal Q−Q
Standardized residuals
Theoretical Quantiles
18
40
42
Scale−Location
√|Standardized residuals|
Fitted values
18
42
40
Residuals vs Leverage
Standardized residuals
Leverage
18
0.5
22
42
Cook's distance

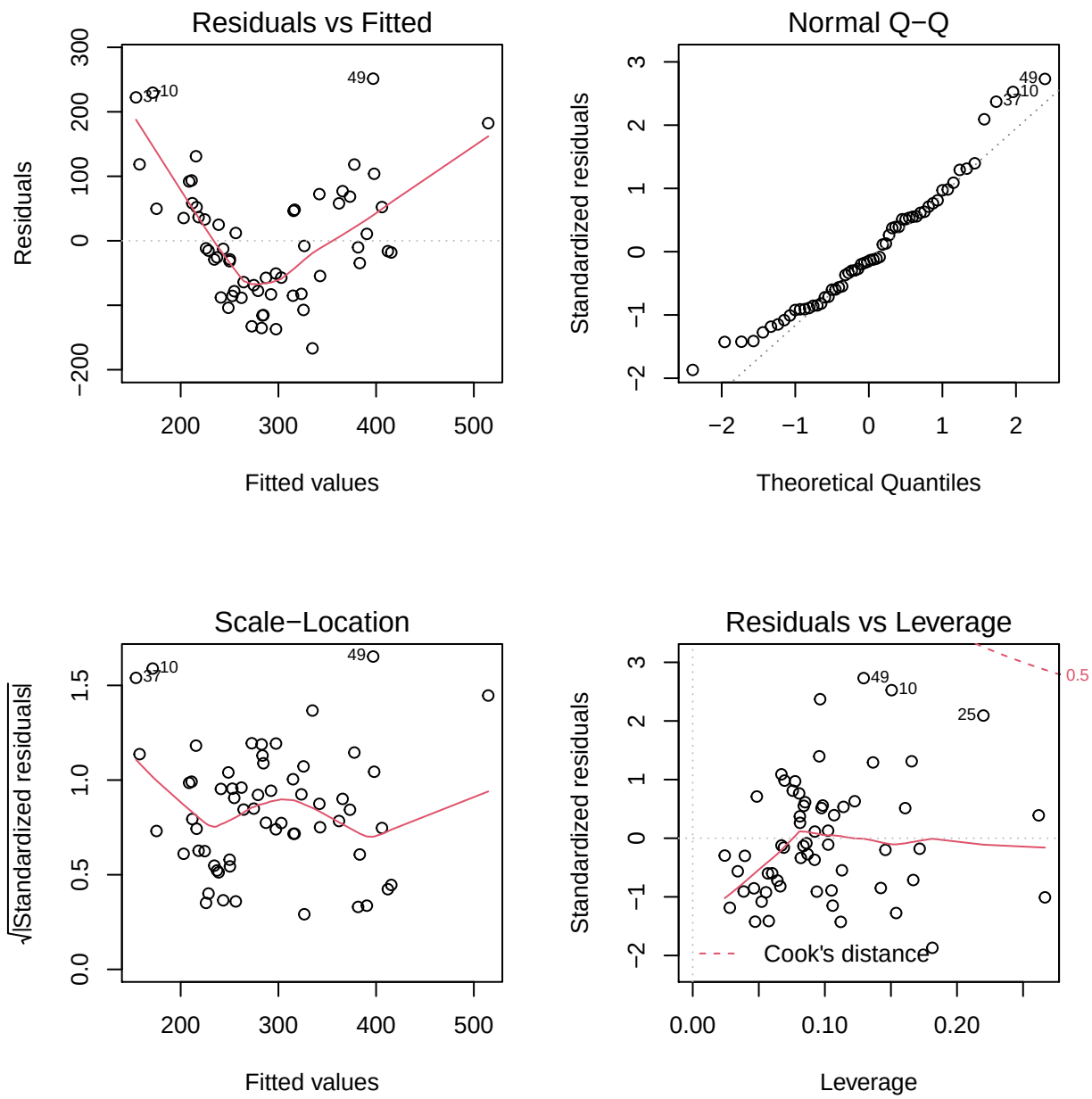
Question 17: Residual Plots - Model 2

Here are the plots for model 2, which includes 80 subjects.



Question 17: Residual Plots - Model 3

Here are the plots for model 3, which includes 60 subjects.



17 Question 17 (6 points)

For each of the three models described above, specify the most appropriate conclusion regarding its residual plots. For each model (1, 2 and 3), you are asked to select one of the following responses...

a. This model has a serious problem with the assumption of linearity

b. This model has a serious problem with the assumption of constant variance
c. This model has a serious problem with the assumption of Normality
d. There are no serious problems evident in this model's residual plots

17.1 Answer 17: Model 1 is c, Model 2 is b and Model 3 is a

These results come from the residual plots. In Model 1, we have a highly influential outlier, calling the assumption of Normality into question. In Model 2, we have clear evidence of a problem with non-constant variance indicated by the rising trend in the scale-location plot. In Model 3, we have clear indications of non-linearity shown by the curve in the residuals vs. fitted values plot.

18 Question 18

The TAs and Dr. Love prepared an analysis to repeatedly monitor accumulated results as they emerged to see if a threshold that indicated that a treatment was beneficial had been crossed. Our approach was designed to deal with a theoretical problem which David Spiegelhalter refers to in The Art of Statistics as the "Law of the Iterated Logarithm." Which of the following describes what we did in an appropriate way? (CHECK ALL RESPONSES THAT APPLY.)

a. performing a sequential probability ratio test
b. avoiding the potential for Simpson's paradox
c. using a threshold to simultaneously control Type I and Type II error
d. accounting for multiple comparisons with a Bonferroni procedure
e. None of the above.

18.1 Answer 18 requires both a and c.

See Chapter 10 of Spiegelhalter's *The Art of Statistics*.

19 Question 19 (4 points)

Once a confidence interval is calculated, several design changes may be used by a researcher to make a confidence interval wider or narrower. For each of the changes listed below, indicate the impact on the width of the confidence interval.

- Rows
  a. Increase the sample size.
  b. Increase the level of confidence.
  c. Increase the standard error of the estimate.
  d. Use a bootstrap approach to estimate the CI.
- Columns
  1. CI will become wider
  2. CI will become narrower
  3. CI width will not change
  4. It is impossible to tell

19.1 Answers for 19 are a = 2, b = 1, c = 1, d = 4

Increasing the sample size will decrease the standard error, and thus decrease the width of the confidence interval.

Increasing the level of confidence, or increasing the standard error, will increase the width of the confidence interval.

If we use a bootstrap, our interval might get wider and it might get narrower, so we cannot tell.

20 Question 20

Suppose you have a tibble with two variables. One is a factor called Pollution with levels High, Low and Medium, arranged in that order, and the other is a quantitative outcome called Response. You want to rearrange the order of the Pollution variable so that you can then use it to identify for `ggplot2` a way to split a histogram of Response values into a series of smaller plots, each containing the histogram for subjects with a particular level of exposure (Low then Medium then High.)

Which of the pairs of `tidyverse` functions identified below would be an effective approach to use in developing such a plot?

a. `fct_reorder` and `facet_wrap`
b. `fct_relevel` and `facet_wrap`
c. `fct_collapse` and `facet_wrap`
d. `fct_reorder` and `group_by`
e. `fct_collapse` and `group_by`

20.1 Answer 20 is b

`fct_relevel` lets you specify a new order for factor levels "by hand" which is what you'd need to do here, and `facet_wrap` is the easier approach to getting the individual subsetted histograms in this context.

Choice a is probably going to be popular, too, but `fct_reorder` lets you reorder the factor levels by sorting along another variable's values, and that's not what we're doing here.

21 Question 21

According to Jeff Leek in *The Elements of Data Analytic Style*, which of the following are good reasons to create graphs for data exploration? (CHECK ALL RESPONSES THAT APPLY.)

a. To understand properties of the data.
b. To inspect qualitative features of the data rather than a huge table of raw data.
c. To discover new patterns or associations.
d. To consider whether transformations may be of use.
e. To look for statistical significance without first exploring the data.

21.1 Answer 21: a and b and c and d (all but e)

If this isn't clear, look at Chapter 5 of the book.

Setup for Questions 22-24

Consider the `weather_check` data frame within the `fivethirtyeight` package. We will use these data for Questions 22-24.

Suppose you want to build a table containing information from the `female`, `ck_weather` and `age` variables in that data frame. I suggest you use the following approach to place the data in the `wc` tibble, and adjust some of the coding. We'll take `female` to represent gender identification here.

Note I have provided this code snippet to you in a file called `wc_code.R`

```
wc <- fivethirtyeight::weather_check %>%
  select(female, ck_weather, age) %>%
  mutate(female = fct_recode(factor(female),
                             "Female" = "TRUE",
                             "Male" = "FALSE"),
         ck_weather = fct_recode(factor(ck_weather),
                            "Check" = "TRUE",
                            "No Check" = "FALSE")) %>%
  mutate(female = fct_relevel(female, "Female"),
         ck_weather = fct_relevel(ck_weather, "Check"))
```

22 Question 22

Build the specified table using your `wc` tibble. Which age group has exactly 103 respondents who identify as male who answered yes to the question "Do you typically check a daily weather report?"

a. Ages 18-29
b. Ages 30-44
c. Ages 45-59
d. Ages 60+
e. None of these.

22.1 Answer 22 is d

Here's the table.

```
wc <- fivethirtyeight::weather_check %>%
  select(female, ck_weather, age) %>%
  mutate(female = fct_recode(factor(female),
                             "Female" = "TRUE",
                             "Male" = "FALSE"),
         ck_weather = fct_recode(factor(ck_weather),
                            "Check" = "TRUE",
                            "No Check" = "FALSE")) %>%
  mutate(female = fct_relevel(female, "Female"),
         ck_weather = fct_relevel(ck_weather, "Check"))

wc_table <- wc %$% table(female, ck_weather, age)

ftable(wc_table)
```

```
                  age 18 - 29 30 - 44 45 - 59 60+
female ck_weather
```

```
Female Check              79     105     115 121
       No Check           30      28      29  20
Male   Check              41      56     119 103
       No Check           26      15      15  14
```

23 Question 23

Perform an appropriate test to see if the odds ratio for a Yes answer to "Do you typically check a daily weather report?" comparing respondents who identify as Female to those who identify as Male is essentially consistent across age categories. What is the name of the test that you ran, and what is the conclusion? Use a 5% significance level.

a. Chi-square test on a 2x2 table with `Epi` package's `twoby2` and found a detectable association.
b. Chi-square test on a 2x2 table with `Epi` package's `twoby2` and found NO detectable association.
c. Woolf's test from the `vcd` package, and odds ratio is sufficiently consistent across age categories to let me collapse on age.
d. Woolf's test from the `vcd` package, and odds ratio is NOT sufficiently consistent across age categories to let me collapse on age.
e. None of these statements describes an appropriate test.

23.1 Answer 23 is c

We need to run a Woolf test on the table we just built. At the 5% level, we cannot identify a detectable difference in the odds ratios, so our conclusion matches c.

```r
vcd::woolf_test(wc_table)
```

```
    Woolf-test on Homogeneity of Odds Ratios (no 3-Way assoc.)

data:  wc_table
X-squared = 6.5613, df = 3, p-value = 0.08728
```

24 Question 24

Use the data we have been working with in the previous two questions, regardless of how you answered those questions. Suppose we want to use the Cochran-Mantel-Haenszel approach to estimate the common odds ratio across all age categories comparing the two gender groups as to whether they check the weather daily. Which of the following statements is true?

a. Females have higher odds in the sample of checking the weather, and a 95% confidence interval includes 1.
b. Females have higher odds in the sample of checking the weather, and a 95% confidence interval does not include 1.
c. Males have higher odds in the sample of checking the weather, and a 95% confidence interval includes 1.
d. Males have higher odds in the sample of checking the weather, and a 95% confidence interval does not include 1.
e. None of these statements are accurate.

24.1 Answer 24 is c

Now we will run the Cochran-Mantel-Haenszel test. The 95% CI for the odds ratio ranges from (0.65, 1.29) and the point estimate for the odds of a weather check given female (as compared to male) is 0.92, so males have higher odds, but the CI includes 1.

```
mantelhaen.test(wc_table)
```

```
    Mantel-Haenszel chi-squared test with continuity correction

data:  wc_table
Mantel-Haenszel X-squared = 0.17397, df = 1, p-value = 0.6766
alternative hypothesis: true common odds ratio is not equal to 1
95 percent confidence interval:
 0.6533502 1.2870384
sample estimates:
common odds ratio
        0.9169988
```

25 Question 25

According to Jeff Leek in *The Elements of Data Analytic Style*, which of the following are good ideas in creating graphs you will share with other people to describe your work? (CHECK ALL RESPONSES THAT APPLY!)

a. If you have multiple plots to compare, use the same scale on the vertical axis.
b. Axis labels should be large, easy to read, in plain language.
c. Add a third dimension, perhaps with animation.
d. Include units in figure labels and legends.
e. Use color and size to help communicate information, for instance to point out confounding.

25.1 Answer 25 is a and b and d and e (all but c)

If this isn't clear, take a look at Chapter 10.

Setup for Questions 26-28

Suppose you fit four candidate regression models to predict the natural logarithm of a measure of predatory behavior in leopards. The four models are nested, in that D is a proper subset of C, which is a proper subset of B, which is a proper subset of A. Specifically, Model A contains seven predictors, Model B contains four of those seven predictors, and Model C contains three of the four Model B predictors, while Model D contains two of the predictors in Model C. You obtain the results shown below.

```
modelA <- lm(log(predatory.behavior) ~
               x1 + x2 + x3 + x4 + x5 + x6 + x7, data = leopard)
modelB <- lm(log(predatory.behavior) ~
               x1 + x2 + x3 + x4, data = leopard)
modelC <- lm(log(predatory.behavior) ~
               x1 + x2 + x3, data = leopard)
modelD <- lm(log(predatory.behavior) ~
               x1 + x2 , data = leopard)
```

```r
res26 <- bind_rows(glance(modelA), glance(modelB), glance(modelC), glance(modelD)) %>%
  mutate(model_name = c("A", "B", "C", "D"))

res26 %>% select(model_name, AIC, BIC, df, df.residual, nobs)
```

```
# A tibble: 4 x 6
  model_name   AIC   BIC    df df.residual  nobs
  <chr>      <dbl> <dbl> <dbl>       <int> <int>
1 A          3763. 3801.     7         492   500
2 B          3759. 3784.     4         495   500
3 C          4161. 4182.     3         496   500
4 D          4164. 4181.     2         497   500
```

26 Question 26

Which of these models does the output suggest will be the best choice to predict the natural logarithm of the predatory behavior measure?

a. Model A
b. Model B
c. Model C
d. Model D
e. The output doesn't suggest a "best" choice

26.1 Answer 26 is b

Of these four models, Model B has the lowest AIC and BIC.

27 Question 27

The following output relates to `modelA` described in the previous question.

```r
round(car::vif(modelA),3)
```

```
    x1     x2     x3     x4     x5     x6     x7
 1.009  1.122  1.138  1.010  1.010 12.269 12.257
```

Which of the following statements is the best conclusion from the `vif` output?

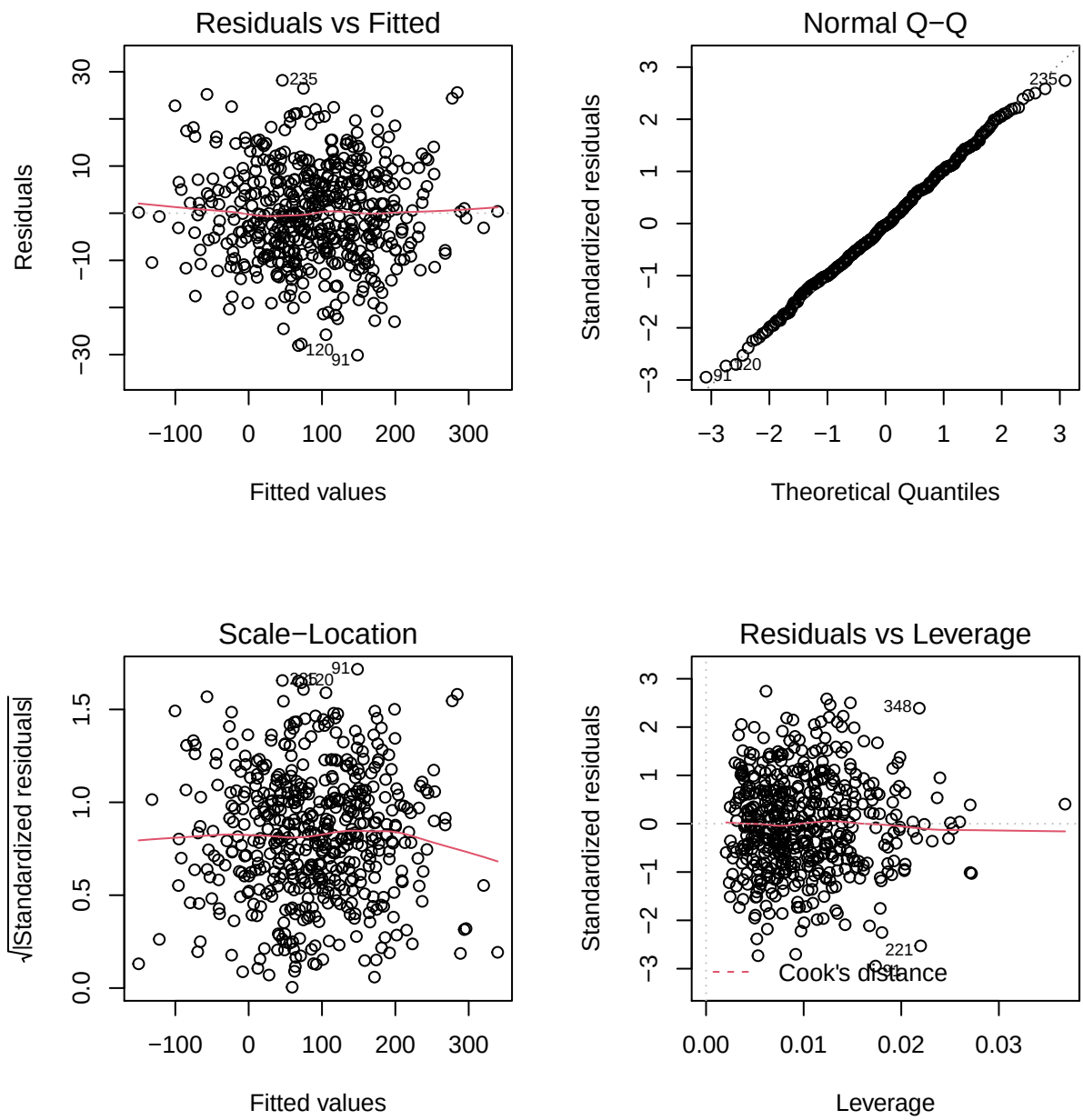
a. Model A's residuals will show no problem with independence
b. Model A's residuals will show a serious problem with independence
c. Model A has no sign of meaningful collinearity
d. Model A has a serious problem with collinearity
e. None of these conclusions is appropriate

27.1 Answer 27 is d

Variables x6 and x7 show signs of substantial variance inflation, due to collinearity, since their VIF values exceed 5. Model A includes these predictors, so it has a collinearity problem.

28 Question 28

Behold the residual plots for Model B from Question 26.



Which of the following conclusions is most appropriate for model B based on the plots provided above?

a. There is a serious problem with the assumption of linearity.
b. There is a serious problem with the assumption of Normality.
c. There is a serious problem with the assumption of constant variance
d. There are no serious problems evident in these residual plots.
e. None of these conclusions are appropriate.

28.1 Answer 28 is d

I see no problems with the assumptions of linear regression evident in these residual plots.

29 Question 29

You are comparing two regression models for the same outcome, which you built using a training sample of data. You then use each model to predict data in a test sample, that was not used to calculate the original regression equations. Which of the following summaries will be useful to you in assessing which model does a better job of out-of-sample prediction? (CHECK ALL RESPONSES THAT APPLY.)

a. The square root of the mean of the squared prediction errors
b. The mean of the absolute values of the prediction errors
c. The maximum of the absolute values of the prediction errors
d. The AIC statistic calculated in the training sample
e. None of these measures would be useful to you

29.1 Answer 29 is a and b and c

All of these are useful except the AIC in the training sample, which is a measure of performance within the sample, but doesn't tell us anything about predicting out of sample.

30 Question 30 (8 points)

Write a clear and well-composed essay of 150 to 200 words describing an important idea from David Spiegelhalter's *The Art of Statistics* about doing statistical science well that Dr. Love didn't cover in Classes 1-24. Your essay should state the idea in your own words, and should indicate why you feel it is important.

If you quote Spiegelhalter (and we prefer that you do), specify the Chapter containing your quote. If Dr. Love discussed your idea in class, you'll lose 1 of 8 available points. If your essay is unclear, or if you miss Spiegelhalter's point, that will have a bigger impact on your score. Each Chapter in Spiegelhalter includes a summary of key points. Feel free to use these summaries to help spark ideas, but do not quote the summaries.

Use any software you like to produce a Word document or PDF submission.

These instructions are 150 words long.

30.1 Answer 30 is an essay.

We don't write sketches for essay questions.

Dr. Love reviewed the essays with regard to the following issues:

1. Did the essay meet the general specifications, regarding length and topic?
2. Were there multiple problems with grammar/syntax/spelling?
3. Was the topic of the essay clear to me on a single reading?
4. Was the topic something that is discussed in The Art of Statistics?
5. Had we discussed the topic of the essay meaningfully in class?
6. What was the overall quality of the essay?

Assuming items 1-4 went fairly well, my reactions to items 5 and 6 led me to your grade. If items 1-4 above didn't go so well, then you lost somewhere between 0.5 and 3 points for those concerns.

- I didn't actually give a score of 8 out of 8, but I also didn't spread out the scores too much, even though the responses were incredibly wide-ranging.
- About 1/4 of the class scored 7.5 out of 8, which was usually a very good essay about a topic we hadn't really discussed, or a really excellent essay about something we had discussed.
- About 1/3 of the class scored 7 out of 8, which was usually a good essay about a topic we hadn't really discussed, or a very good essay about something we had discussed.
- About 1/4 of the class scored 6.5 out of 8, which was usually an OK essay about a topic we hadn't really discussed, or a good essay about something we had discussed.
- The rest of the class scored below 6.5, which usually indicated some problems with items 1-4 above, or that the essay itself just wasn't very effective or clear.
- Sorry I cannot provide more detailed feedback, but most of the essays were at least OK, which was good news. The mean score on the essay was 85% of available credit.

Alphabetically, the essay topics were (as summarized by me):

- Algorithmic Transparency (2)
- Ascertainment Bias
- Association vs Causation (4)
- Bayes factors
- Bayes' Theorem
- Bayesian "smoothing"
- Bias/Variance Tradeoff
- Bradford-Hill Criteria (2)
- Causality and Data Reversibility
- Central Limit Theorem
- Classification Trees (2)
- Communication & Transparency (5)
- Communication Pipeline (3)
- Complexity vs Performance
- Confounding
- Convenience Sampling
- Data Ethics
- Designing Observational Studies
- Ethics / Statistical Fraud
- Evaluating Algorithms (3)
- Evaluating Published Claims (2)
- Exploration vs. Confirmation
- External Validity
- Feature Engineering (2)
- Making Daily Decisions
- Metaphorical Populations
- Mixing Fisher and Neyman-Pearson
- Neural Networks
- Over-fitting
- P-hacking
- Poisson Distribution (2)
- Positive vs. Negative Framing
- PPDAC cycle
- Pre-registration of analyses
- Publication Bias (3)
- Questionable Research Practices / Researcher Degrees of Freedom (7)
- Randomized Clinical Trials
- Regression to the Mean (2)
- Reproducibility Crisis (4)
- ROC curves

- Simpson's Paradox

That's a handy index of things I would hope you would have a better understanding of after reading Spiegelhalter.

31 Grading Results

- I awarded partial credit whenever I could justify it.
- The items describing the survey (items 1-15) that went poorly included 6a, 8b, 9a and 9b, 11a and 11b, 12a and 12b and 15a. In general, those were meant to be the more difficult (or at least more exacting) items, so that's OK.

Item	% with Correct Response	% of Points Awarded	Available Points
1a	70	70	1.5
1b	80	80	1.5
2a	100	100	1.5
2b	93	96	1.5
3a	93	96	1.5
3b	76	90	1.5
4a	96	96	1.5
4b	96	96	1.5
5a	96	97	1.5
5b	79	81	1.5
6a	17	45	1.5
6b	86	86	1.5
7a	89	89	1.5
7b	91	91	1.5
8a	94	94	1.5
8b	36	36	1.5
9a	40	59	1.5
9b	67	68	1.5
10a	97	97	1.5
10b	96	96	1.5
11a	57	59	1.5
11b	59	59	1.5
12a	67	77	1.5
12b	64	76	1.5
13a	96	96	1.5
13b	76	86	1.5
14a	81	85	1.5
14b	87	87	1.5
15a	59	76	1.5
15b	81	81	1.5

- The items that went poorly in the rest of the Quiz were 16a and 16b, 17 (Model 1), 23 and 24. None of those was a really big surprise to me, either.

Item	% with Correct Response	% of Points Awarded	Available Points
16a	37	40	1.5
16b	43	50	2.5
17_1	39	39	2
17_2	97	97	2

Item	% with Correct Response	% of Points Awarded	Available Points
17_3	94	94	2
18	40	80	3
19a	96	96	1
19b	93	93	1
19c	90	90	1
19d	80	80	1
20	90	90	3
21	40	80	3
22	93	93	3
23	53	53	3
24	63	63	3
25	86	95	3
26	63	63	3
27	99	99	3
28	94	94	3
29	73	86	3
30	0	85	8

The highest score a student earned on Quiz 2 was 95 out of 100. If I was assigning letter grades solely based on this Quiz I would require at least 85% for some form of A and 70% for some form of B.

- 12 grades were between 90 and 95
- 17 grades were between 85 and 89.75
- 16 grades were between 80 and 84.75
- 12 grades were between 75 and 79.75
- 7 grades were between 70 and 74.75
- 6 students scored below 70

32 System Information

```r
sessionInfo()
```

```
R version 4.0.2 (2020-06-22)
Platform: x86_64-w64-mingw32/x64 (64-bit)
Running under: Windows 10 x64 (build 19041)

Matrix products: default

locale:
[1] LC_COLLATE=English_United States.1252
[2] LC_CTYPE=English_United States.1252
[3] LC_MONETARY=English_United States.1252
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.1252

attached base packages:
[1] grid      stats     graphics  grDevices utils     datasets  methods
[8] base

other attached packages:
```

```
 [1] forcats_0.5.0     stringr_1.4.0     purrr_0.3.4       readr_1.4.0
 [5] tidyr_1.1.2       tibble_3.0.4      tidyverse_1.3.0   broom_0.7.2
 [9] Hmisc_4.4-2       Formula_1.2-4     survival_3.2-7    Epi_2.42
[13] vcd_1.4-8         patchwork_1.1.0   magrittr_2.0.1    car_3.0-10
[17] carData_3.0-4     ggrepel_0.8.2     naniar_0.6.0      mosaic_1.8.2
[21] ggridges_0.5.2    mosaicData_0.20.1 ggformula_0.9.4   ggstance_0.3.4
[25] dplyr_1.0.2       Matrix_1.2-18     ggplot2_3.3.2     lattice_0.20-41
[29] janitor_2.0.1     knitr_1.30

loaded via a namespace (and not attached):
 [1] colorspace_2.0-0      ellipsis_0.3.1        rio_0.5.16
 [4] visdat_0.5.3          leaflet_2.0.3         snakecase_0.11.0
 [7] htmlTable_2.1.0       fs_1.5.0              base64enc_0.1-3
[10] ggdendro_0.1.22       rstudioapi_0.13       farver_2.0.3
[13] fansi_0.4.1           lubridate_1.7.9.2     xml2_1.3.2
[16] splines_4.0.2         polyclip_1.10-0       jsonlite_1.7.1
[19] cluster_2.1.0         dbplyr_2.0.0          png_0.1-7
[22] fivethirtyeight_0.6.1 ggforce_0.3.2         compiler_4.0.2
[25] httr_1.4.2            backports_1.2.0       assertthat_0.2.1
[28] cli_2.2.0             tweenr_1.0.1          htmltools_0.5.0
[31] tools_4.0.2           gtable_0.3.0          glue_1.4.2
[34] Rcpp_1.0.5            cellranger_1.1.0      vctrs_0.3.5
[37] nlme_3.1-148          crosstalk_1.1.0.1     lmtest_0.9-38
[40] xfun_0.19             rvest_0.3.6           openxlsx_4.2.3
[43] lifecycle_0.2.0       mosaicCore_0.8.0      MASS_7.3-53
[46] zoo_1.8-8             scales_1.1.1          hms_0.5.3
[49] parallel_4.0.2        RColorBrewer_1.1-2    yaml_2.2.1
[52] curl_4.3              gridExtra_2.3         rpart_4.1-15
[55] latticeExtra_0.6-29   stringi_1.5.3         checkmate_2.0.0
[58] zip_2.1.1             rlang_0.4.9           pkgconfig_2.0.3
[61] evaluate_0.14         htmlwidgets_1.5.2     cmprsk_2.2-10
[64] tidyselect_1.1.0      plyr_1.8.6            R6_2.5.0
[67] generics_0.1.0        DBI_1.1.0             pillar_1.4.7
[70] haven_2.3.1           foreign_0.8-80        withr_2.3.0
[73] mgcv_1.8-31           abind_1.4-5           nnet_7.3-14
[76] etm_1.1.1             modelr_0.1.8          crayon_1.3.4
[79] utf8_1.1.4            rmarkdown_2.5         jpeg_0.1-8.1
[82] readxl_1.3.1          data.table_1.13.2     reprex_0.3.0
[85] digest_0.6.27         numDeriv_2016.8-1.1   munsell_0.5.0
```